What is proof?

A mathematical proof is an inferential argument for a mathematical statement, showing that stated assumptions logically guarantee the conclusion. The argument may use other previously established statements, such as theorems but every proof can, in principle, be constructed using only certain basic or original assumptions known as axioms, along with the accepted rules of inference

Different types of proofs

- Direct proof
- Proof by Mathematical Induction
- Proof by Contraposition
- Proof by Contradiction
- Proof by Construction
- Proof by exhaustion
- Probabilistic proof
- Combinatorial proof
- Non constructive proof
- Statistical proofs in pure math
- Computer - assisted proofs.

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Proof by Contraposition

Proof by contraposition infers the statement "if p then q" by establishing the logically equivalent contrapositive statement "if not p then not q"

Example

Prove that if x is even then x^2 is even

Suppose that x is not even. The product of two odd numbers is odd, hence $x^2 = x \cdot x$ is odd. Thus x^2 is not even. Thus if x^2 is even, the supposition, must be false, so x has to be even

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Application

- Used in Financial industries (Once used to uncovex a Ponzi scheme)
- Used in Stock Market.
- Used in A.I.
- Used to verify the correctness of a statement

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